1. Consider the function $f(x) = \frac{2}{\sqrt{x^2-5}}$

A. Find the Linearization of the function at a = 3.

B. Using a plotter, find where the function approximates the function to within a tolerance of 0.1

2. Find the differential, dy, of

A. $y = \frac{1+2x}{1+3x}$

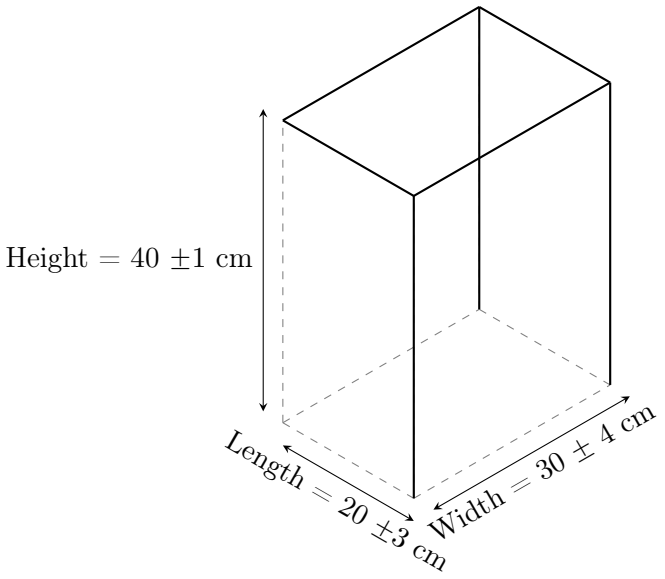
B. $y = x^2 \sin(2x)$

3. One side of a right triangle is known to be 20 cm long and the opposite angle is measured as 30°, with a possible error of 1°.

A. Use differentials to estimate the error in computing the length of the hypotenuse.

B. What is the percentage error?

4. The length, width, and height of the box are all known to some uncertainty. (l,w,h) = (20 ±3 cm, 30 ±4 cm,40 ±1 cm).



A. What is the fractional uncertainty of each dimension of the box

B. What is fractional uncertainty of volume?

C. What is the uncertainty of the volume?